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LIFT AND PITCHING-MOMENT INTERFERENCE BETWEEN A POINTED CYLINDRICAL BODY AND TRIANGULAR WINGS OF VARIOUS ASPECT RATIOS AT MACH NUMBERS OF 1.50 AND 2.02

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The results for the components alone and in combination were used to determine the total interference, which is defined as the sum of the interference effects of the body on the wing forces and of the wings on the body forces. The interference effects were important for the wing-body combinations having small wings relative to the body. Both the results of the theory of NACA Rep. 962 and of the modified theory were in good agreement with the experimentally measured interference results.

INTRODUCTION

The forces on a combination of a wing and a body can be considered to consist of the sum of the forces on the wing alone, the body alone, and the interference forces of the wing on the body and of the body on the wing. Several investigators have presented theoretical methods of predicting interference forces. Spreiter, in reference 1, has investigated the effect of interference on the lift-curve slope and center-of-pressure position of slender wing-body combinations. This theory assumes that the body is slender and the leading edges of the wings are swept well behind the Mach cone. Ferrari, in reference 2, has investigated the problem of interference between a rectangular wing and a body. In this paper the effect of the wing on the body forces, assuming that the flow field due to the wing is unchanged by the presence of the body, and the effect of the body on the wing forces, assuming that the body flow field is unchanged by the presence of the wing, were determined. Brown, Friedman, and Hodes, in reference 3, have investigated the conical-flow problem of interference between a triangular wing and a conical body, the apex of which coincides with the wing apex.

The present experiments were designed to measure the total lift and pitching-moment interference of triangular wing-body combinations at supersonic speeds and to compare the data with the theory and a modification of the theory of reference 1. The experiments also afforded an opportunity for comparison of the lift force and pitching moment of the body and wings alone with values predicted by the available theories. The total interference, which is defined as the sum of the interference effects of the body on the wing forces and of the wing on the body forces, was determined by substracting the sum of the lift, or pitching moment, of the wings and body alone from the lift, or pitching moment, of the corresponding combinations.

NOTATION

A	wing aspect ratio
A_p	plan-form area of body $\left(2\int_0^l a\,dx\right)$, sq in.
a	local body radius, in.

$\bar{c}$ mean aerodynamic chord $\left(\frac{2}{3} c_r\right)$, in.

c_{d_c} cross-flow section drag coefficient of a circular cylinder

C_L lift coefficient based on total wing plan-form area for wings and combinations and on base area for body

ΔC_{L_s} increment in lift coefficient due to stream angle

C_m pitching-moment coefficient about wing centroid for wings and combinations and about body nose for body, based on total wing plan-form area and mean aerodynamic chord for wings and combinations, and on base area and body length for body

ΔC_{m_s} increment in moment coefficient due to stream angle

c_r wing apex chord, in.

E complete elliptic integral of second kind

L lift force, lb

l body length, in.

$\frac{L_{WB}+L_{BW}}{L_B+L_W}$ total lift-interference ratio $\left(\frac{L_{WB}+L_{BW}}{L_B+L_W} = \frac{L_C}{L_B+L_W} - 1\right)$

M pitching moment, in.-lb

M_o free-stream Mach number

$\frac{M_{WB}+M_{BW}}{M_B+M_W}$ total moment-interference ratio, moments about body nose $\left(\frac{M_{WB}+M_{BW}}{M_B+M_W} = \frac{M_C}{M_B+M_W} - 1\right)$

$\frac{\Delta p}{q}$ loading coefficient, ratio of difference between lower- and upper-surface static pressures and free-stream dynamic pressure

s local wing semispan, in.

S total wing plan-form area as extended in figure 1 ($S = c_r s_m$), sq in.

V volume of body, cu in.

V_o	free-stream velocity, in./sec
x	longitudinal coordinate, measured along body axis from body nose for body alone and combination, or measured along wing apex chord from wing apex for wings, positive downstream, in.
y	lateral coordinate, normal to vertical plane of symmetry, in.
α	angle of attack in radians unless otherwise specified
α_s	stream angle, radians
β	$\sqrt{M_o^2 - 1}$
ϵ	wing semiapex angle, deg
λ	modification factor to account for finite wing aspect ratios
η	correction for three-dimensional effects on body
Λ_o	sweep angle of wing leading edge, deg
$\Lambda_{1/2}$	sweep angle of wing midchord line, deg
φ	velocity potential

Subscripts

B	body alone
W	wing alone
C	wing-body combination
WB	effect of wing on body
BW	effect of body on wing
$L \rightarrow 0$	limiting value of quantity as lift approaches zero
b	value at body base
l	value at intersection of wing leading edge and body
m	maximum value
s	value due to stream angle
t	value at the wing trailing edge

∞	theoretical value for infinite aspect ratio
bc	centroid of body plan-form area
cp	center of pressure of wing-body combination

EXPERIMENTAL CONSIDERATIONS

Apparatus and Procedure

The tests were performed in the Ames 1- by 3-foot supersonic wind tunnel No. 1. This closed-circuit continuous-operation wind tunnel is equipped with a flexible-plate nozzle that can be adjusted to give test-section Mach numbers from 1.2 to 2.4. Reynolds number variation is accomplished by changing the absolute pressure in the tunnel from one-fifth of an atmosphere to approximately three atmospheres depending on the Mach number and ambient temperature. The tunnel is equipped with a strain-gage balance for measuring the aerodynamic forces on sting-supported models (ref. 4). In the arrangement described in reference 4, the pitching moment was obtained from the reactions on the main balance springs and was not sufficiently accurate. Therefore, the pitching moment in the present investigation was more accurately determined from strain-gage measurements of the bending moment in the sting support (ref. 5).

The models were tested through a nominal angle-of-attack range of $\pm 5.5^{\circ}$ at Mach numbers of 1.50 and 2.02. A constant Reynolds number of 0.5 million per inch was maintained and, in order to make the effects of condensation negligible, the humidity was held to less than 0.0003 pound of water vapor per pound of dry air.

Models and Supports

The body (fig. 1) had a fineness ratio of 7.33, a conical nose with a semiapex angle of 15°, and an ogival transition section fairing into a cylindrical afterbody. The length of the body was limited by the condition that the nose wave reflected from the tunnel side walls should fall behind the body base.

The geometrical properties and designations of the six wing models used in the investigation are summarized in table I. A photograph of the wing family is presented in figure 2. The wings had symmetrical double-wedge airfoil sections in the streamwise direction with a maximum thickness of 8 percent at the midchord. All the wings were made of hardened tool steel and were finished by grinding. They were all equipped with small supports which were designed to reduce the effect of the supports on the aerodynamic forces of the wing alone to a negligible quantity.

For all the wing-body combinations the wings were located along the cylindrical part of the body. The method of assembling the combinations is shown in figure 3.

All the models were mounted on the same sting. However, as shown in figure 4, different shrouds were used for the wing tests than for the body and combination tests.

ANALYSIS OF DATA

Corrections to Experimental Results

The experimental lift and moment data have been corrected for the nonuniform flow conditions in the tunnel test section. The measured values of the stream angle and pressure coefficient in the vertical plane of symmetry of the empty tunnel were used, together with the theoretical results of the appendix, in estimating the corrections. It was found, in general, that the corrections to lift and moment were small but not entirely negligible. The maximum correction to lift-curve slope for all configurations at both Mach numbers was 10 percent of the measured lift-curve slope. The corrections to the moment data, at both Mach numbers, shifted the center of pressure of the body 4 percent of the body length; the center of pressure of the wings, a maximum of 3 percent of the wing mean aerodynamic chord; and the center of pressure of the wing-body combinations, a maximum of 3 percent of the body length.

Precision

The precision of the experimental data has been evaluated by the method outlined in Appendix A of reference 5. This includes an estimate of the precision of each measurement and the resulting uncertainty in the measurement. There is a further uncertainty involved in the accuracy of the corrections applied to the experimental data of the present tests. The latter inaccuracy is estimated to cause an uncertainty of ±0.007 in the lift coefficients for body, wings, and wing-body combinations; an uncertainty of ±0.006 in the moment coefficients for the body and an uncertainty of ±0.004 in the moment coefficients for the wings and the wing-body combinations. The total uncertainty in the results is taken as the square root of the sum of the squares of the individual uncertainties.

The following table lists the total uncertainty for all configurations at both Mach numbers:

Quantity	Uncertainty for body	Uncertainty for wings and wing-body combinations
M_o	±0.02	±0.02
C_L	±.009	±.009
C_m	±.007	±.005
α(deg)	±.10	±.10

THEORETICAL CONSIDERATIONS

Body

Tsien (ref. 6) showed that the lift force and pitching moment on slender bodies of revolution at low angles of attack are the same at supersonic speeds as at subsonic speeds, and that the results are the same as those predicted by Munk's airship theory (ref. 7). Thus, the lift-curve slope of a body with a finite base is 2 for all Mach numbers if the base is used as the reference area. Experiments have shown that, while this is a good approximation at low angles of attack, at higher angles of attack the lift-curve slope increases and the slender-body theory is no longer adequate. Slender-body theory neglects the effects of viscosity and considers only the potential flow about the body. A large effect of viscosity can be included by considering the flow of a real fluid about an infinite cylinder inclined to the stream. In reference 8, Jones has shown that the forces on an inclined infinite cylinder are determined by the cross flow, that is, the component of the flow perpendicular to the cylinder. Since the flow of a real fluid normal to a cylinder usually separates, a drag of cross flow occurs and appears as a normal force on the inclined cylinder. Allen (ref. 9) has estimated the effects of cross-flow separation on the aerodynamic coefficients of slender bodies of revolution. The lift coefficient, by the method of reference 9, is

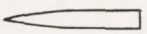
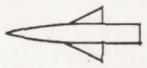
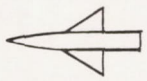
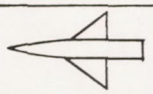
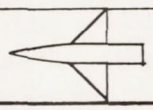
$$C_L = 2\alpha + \frac{A_p}{\pi a_b^2} c_{d_c} \eta \alpha^2 \qquad (1)$$

The first term represents the contribution of slender-body theory. The second term accounts for the added lift due to the cross-flow separation. In the second term c_{d_c} is the drag coefficient experienced by an infinitely long circular cylinder at the Reynolds number and Mach number based upon the diameter of the body and the cross component of the velocity. The factor η allows for the effect of the finite length of the circular cylinder with the assumption that the reduction in drag coefficient for fineness ratio is the same for each element of the cylinder. It is also

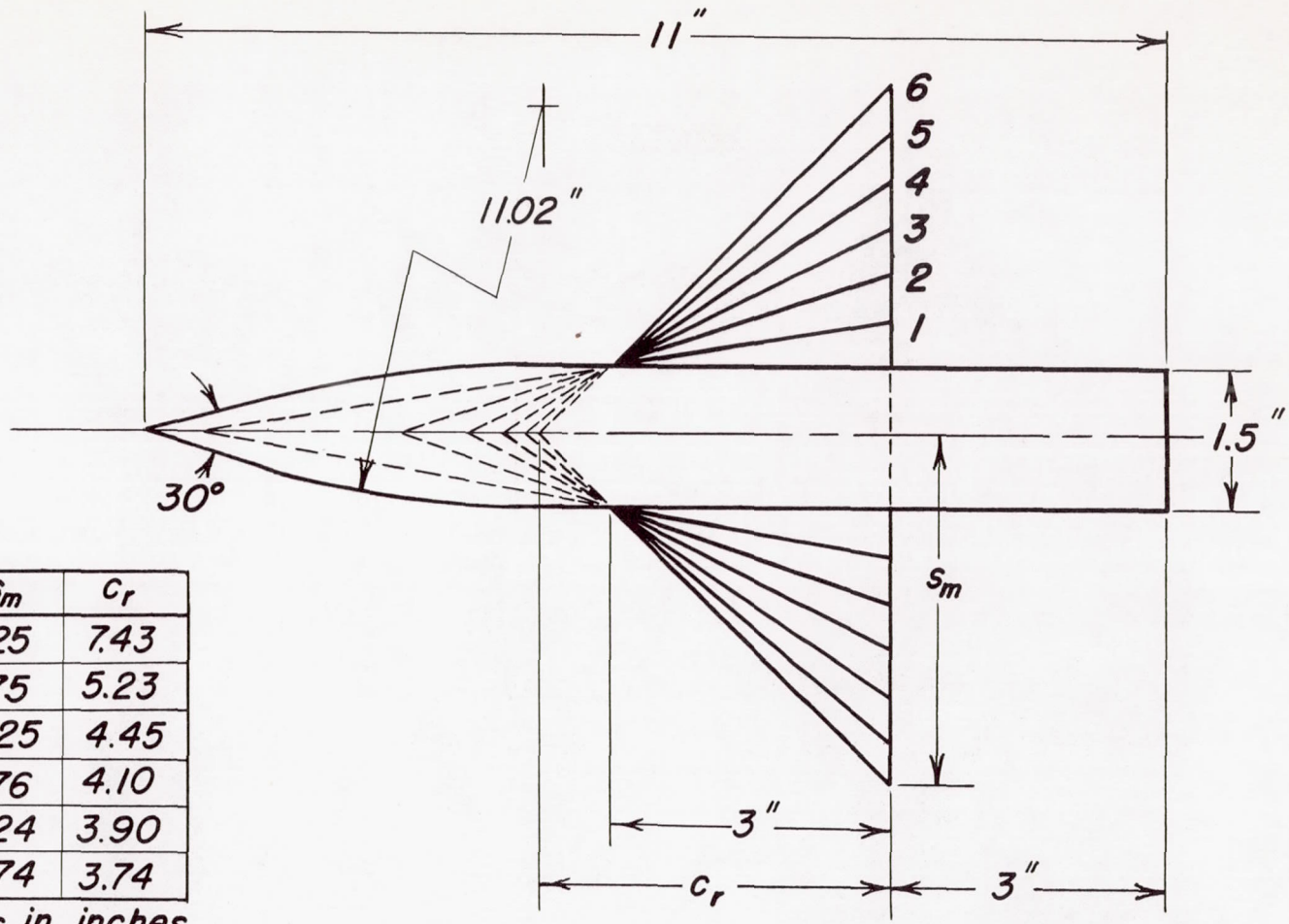
TABLE I.- SUMMARY OF GEOMETRICAL PROPERTIES OF WINGS

Wing	W_1	W_2	W_3	W_4	W_5	W_6
Sketch						
Λ_o (deg)	80.4	71.6	63.2	56.0	50.3	45.0
$\Lambda_{\frac{1}{2}}$ (deg)	71.4	56.2	44.7	36.6	31.0	26.6
s_m (in.)	1.25	1.75	2.25	2.76	3.24	3.74
$\bar{c}$ (in.)	4.95	3.49	2.97	2.73	2.60	2.49
c_r (in.)	7.43	5.23	4.45	4.10	3.90	3.74
S ($in.^2$)	9.29	9.15	10.01	11.30	12.66	13.99
A	0.67	1.34	2.02	2.69	3.33	4.00

TABLE II.– SUMMARY OF RESULTS

Configuration		Lift		Moment	
Symbol	Sketch	$\left(\frac{dC_L}{d\alpha}\right)_{L\to 0}$ (per deg)		$\left(\frac{dC_m}{dC_L}\right)_{L\to 0}$	
		M_o=1.50	M_o=2.02	M_o=1.50	M_o=2.02
B		0.0340 (.0349)	0.0460 (.0349)	–0.20 (–.190)	–0.20 (–.190)
W_1		.0208 (.0176)	.0186 (.0169)	–.09 (0)	–.03 (0)
W_2		.0305 (.0323)	.0275 (.0289)	0 (0)	.06 (0)
W_3		.0387 (.0442)	.0347 (.0374)	.03 (0)	.06 (0)
W_4		.0455 (.0533)	.0395 (.0398)	.03 (0)	.04 (0)
W_5		.0507 (.0602)	.0425 (.0398)	.06 (0)	.04 (0)
W_6		.0544 (.0624)	.0416 (.0398)	.07 (0)	.08 (0)
W_1B		.0160 (.0137)	.0163 (.0134)	.12 (.180)	.20 (.191)
W_2B		.0300 (.0282)	.0275 (.0260)	.11 (.147)	.12 (.175)
W_3B		.0405 (.0408)	.0373 (.0354)	.11 (.113)	.10 (.150)
W_4B		.0473 (.0510)	.0415 (.0395)	.09 (.0941)	.08 (.148)
W_5B		.0526 (.0590)	.0451 (.0405)	.06 (.0819)	.06 (.151)
W_6B		.0571 (.0622)	.0460 (.0410)	.08 (.0798)	.08 (.150)

Note: In each case the experimental value is given first and the corresponding theoretical value indicated in parentheses directly below.



WING	s_m	c_r
1	1.25	7.43
2	1.75	5.23
3	2.25	4.45
4	2.76	4.10
5	3.24	3.90
6	3.74	3.74

Dimensions in inches

Figure 1. – Plan-form dimensions of body, wings, and wing-body combinations.